

FSB 101

Instruction: Answer all questions

1. One of the following is not an economic importance of Protista (a) They are good source of minerals (b) They are consumers and important in the food chain (c) Many of the organisms are pathogenic (d) None of the above
2. Prokaryotic cells possess all except one of these characteristics (a) Possession of plasmids (b) Possession of membrane bounded organelles (c) possession of Pilli (d) possession of flagella
3. The continuous and gradual transformation of lines of descents is? (a) evolution (b) mitosis (c) meiosis (d) cell division
4. Examples of structural protein include (a) collagen (b) elatin (c) phospholipids (d) iron
5. Binomial nomenclature in classification consists of two words made up of __ and __ (a) family species (b) class, species (c) order species (d) genus, species
6. The first modern system of classification of living organisms was proposed by a/an ____ (a) botanist (b) mycologist (c) zoologist (d) pharmacist
7. One of the following is a characteristic of the kingdom monera (a) they are eukaryotic organisms (b) nutrition is autotrophic or heterotrophic (c) they are organized nucleus (d) they are not involved in nitrogen fixation
8. The great unity of life shows that organism have the same __ (a) ancestor (b) diversity (c) genetic code (d) blood group
9. Life has existed on earth for how many years? (a) 2.7billion years (b) 1.7billion years (c) 3.7billion years (d) 4.6billion years
10. Transport proteins that permit charged particles to cross the bilayer by diffusion is called? (a) carrier proteins (b) channel proteins (c) transfer proteins (d) none
11. An example of symport movement in cells is (a) glucose uptake (b) exocytosis (c) endocytosis (d) none of the above
12. The role of CO_2 during photosynthesis is? (a) reduction of water (b) reduction of ribulose diphosphate (c) reduction of diphosphoglycerate (d) reduction of oxygen
13. The primary atmosphere of the earth was? (a) oxidation (b) neutral (c) reductive (d) reactive
14. The process by which evolution works on genes, phenotypes and population is called? (a) modern synthesis (b) Darwinism (c) Lamarckism (d) speciation
15. Marasmus is a deficiency that affects children of ages (a) 1 – 3years (b) 3 – 5years (c) 5 – 7years (d) 7 – 9years
16. The building block of nature are (a) proteins (b) DNA (c) amino acid (d) genetic code
17. Conditions that require saprophytism are the following except (a) air pressure (b) presence of water (c) presence of oxygen (d) temperature

18. The neo Darwin theory of evolution (a) recognizes the importance of mutation and variation (b) recognizes natural selection (c) recognizes symbiosis (d) fixed and provided evidence of decent

19. In herbivorous mammals, the space between the molars and the incisors is called (a) dentition space (b) diastema (c) wisdom teeth

20. The insect vector of trypanosomiasis is (a) mosquito (b) sand fly (c) Tse-tse fly (d) Ticks

21. A good parasite has the following qualities except (a) high fecundity (b) tough skin (c) kill its host (d) adapt to host

22. The function of centriole of a cell is (a) formation of cilia and flagella (b) site of protein synthesis (c) site of fat synthesis (d) transport of molecule

23. Amino acids are soluble in (a) water (b) palm oil (c) paints (d) hydrocarbon

24. Green plants synthesize carbohydrate through this mode of nutrition (a) holozoic (b) parasitic (c) autotrophic (d) heterotrophic

25. ___ is an example of open chain structure of carbohydrates (a) (β) -lactose (b) α -glucose (d) D-fructose (d) D-glucose

26. An example of exception to cell theory is (a) onion cell (b) animal cell (c) viral cell (d) fungal cell

27. Example of prokaryotic cells are (a) fungi and eubacteria (b) archaeans and eubacteria (c) algae and archaeans (d) protozoa and moss

28. Carbohydrates can be classified into (a) 2 groups (b) 3 groups (c) 4 groups (d) 5 groups

29. ___ is an example of aromatic amino acid (a) glycine (b) methionine (c) aspartic acid (d) phenylalanine

30. What is the function of mitochondria (a) photolysis (b) oxidative metabolism (c) photophosphorylation (d) carbohydrate synthesis

31. Chromatids first appear during (a) Leptotene (b) pachytene (c) zygotene (d) diplotene

32. Which of the following is not a feature of Diakinesis (a) ribonucleic acid synthesis stops (b) appearance of four distinct chromatids (c) nucleolus disappear (d) nuclear membrane appears

33. Give the phenotypic ratio of the F_1 generation of a cross between Tt (father) and tt (mother) where T is trait for Tall stem and t is trait for short stem (a) 2Tall : 2short (b) 2Tall : 1short (c) 1Tall : 2short (d) all Tall

34. Which of these is not a passive process? (a) osmosis (b) electro-osmosis (c) ionic movement (d) facilitated diffusion

35. Which of the following statement is not correct about meiosis (a) cell division that reduces the chromosome number by half (b) four haploid cells are usually formed from the division of a cell (c) two haploid cells are always formed from the division of a cell (d) it is a reduction division

36. The Calvin cycle of photosynthesis consists of the following phases (a) carboxylation, phosphorylation, reduction and regeneration (b) regeneration, photolysis, oxygenation

and reduction (c) phosphorylation, carboxylation, reduction and photolysis (d) reduction, oxygenation, photolysis and carboxylation

37. The modern man was first discovered in (a) Africa (b) Europe (c) Asia (d) Australia

38. Amino acids contain both amino and (a) sulphate (b) hydrochloric acid (c) carbohydrates (d) carboxyl group

39. The human body cannot biosynthesize ___ (a) essential amino acids (b) hormone (c) pancreatic juice (d) hydrochloric acid

40. The ribosomes of prokaryotic cells are ___ in size (a) 15S units (b) 30S units (c) 50S units (d) 70S units

41. Which of the following statements is not correct? (a) an enzyme active site binds with substrate like a key fitting into a lock (b) an enzyme increases the activation energy for a reaction (c) the substrate molecule does not normally fit exactly in the active site (d) an enzyme can be reused

42. Which of the following statements is correct? (a) viruses multiply by binary fission (b) viruses multiply by cell division (c) viruses multiply by replication (d) viruses multiply by mitosis

43. In mitosis, both daughter cells have the (a) haploid number of chromosomes (b) diploid number of chromosomes (c) half the number of chromosomes in the somatic cells (d) half the number of chromosomes in the sex cells

44. Cell organelles of Eukaryotic cell with double-layered membrane are ___ (a) nucleus, mitochondria, chloroplast (b) ribosomes, chloroplast, centriole (c) ribosomes, golgi apparatus, mitochondria (d) all of the above

45. ___ is an example of simple lipids (a) Tristearia (b) Lectithin (c) Sulfutids (d) Carotenoids

46. One of these is not an example of macronutrients (a) fats (b) fibre (c) protein (d) water

47. Molecules of carbohydrate contain all the following molecule except (a) carbon (b) nitrogen (c) oxygen (d) hydrogen

48. When the products of a reaction are less complex and more disordered than the reactants, then (a) entropy increases (b) entropy decreases (c) enthalpy increases (d) enthalpy decreases

49. Which of the following equations is correct (a) $\Delta H = \Delta G - T\Delta S$ (b) $\Delta G = \Delta H + T\Delta S$ (c) $\Delta G = \Delta H - T\Delta S$ (d) $\Delta S = \Delta T + \Delta H + \Delta G$

50. Hemorrhage can be managed with ___ (a) vitamin B (b) vitamin K (c) fat and oil (d) vitamin D

51. The main function of the glucagon is the ___ (a) formation of glycogen in the muscle (b) transport of glucose to the liver (c) formation of protein from the amino acid (d) breakdown of glycogen in the liver into glucose

52. The smallest level of taxonomy is (a) species (b) family (c) class (d) genus

Pseud

53. Which of the following statements is true? (a) energy can be transformed from one form to another (b) energy increases as disorder increases (c) energy cannot be transformed from one form to another (d) energy decreases as disorder decreases
54. Which of these statements describes the second law of thermodynamics? (a) entropy in the universe is decreasing (b) entropy in the universe is increasing (c) kinetic energy in the universe is increasing (d) potential energy in the universe is increasing
55. The sum total of all biochemical reactions in a cell that maintain its structural and functional integrity (a) entropy (b) metabolism (c) anabolism (d) free energy
56. When ΔG is negative, the reaction is (a) endergonic (b) exothermic (c) exergonic (d) endothermic
57. In an exothermic reaction (a) ΔG is positive (b) ΔS is negative (c) ΔH is negative (d) ΔT is negative
58. The net energy production during glycolysis is (a) 1 ATP molecule (b) 3 ATP molecules (c) 2 ATP molecules (d) 4 ATP molecules
59. The net energy production during tissue respiration is (a) 38 ATP molecules (b) 34 ATP molecules (c) 36 ATP molecules (d) 32 ATP molecules
60. Which of the following statements about mitosis is not correct (a) during prophase, nucleoli begin to disappear (b) appearance of the spindle during metaphase (c) duplicated chromosomes join one another (d) duplicated chromosomes separate from one another
61. Which of the following statements about mitosis is not correct? (a) appearance of the spindle (b) migration of the spindle between the poles (c) doubling of the chromosome number (d) formation of the equatorial plate
62. Meiosis II can also be referred to as (a) heterotypic division (b) equatorial division (c) reduction division (d) haplitic division
63. Which of the stages during meiosis I (first division in meiosis) does synapsis occur (a) Leptoten (b) Pachytene (c) Zygotene (d) Diplotene
64. Jawless fish was predominant during the ___ period (a) Silurian (b) carboniferous (c) Ordovician (d) Permian
65. The scientific name for Java man is (a) *Homo erectus* (b) *Australopithecus Africanus* (c) *Australopithecus robustus* (d) *Homo sapiens*
66. The following are evidence of evolution except (a) horizontal gene transfer (b) intelligent design (c) phylogenetic tree (d) natural selection
67. Dinosaurs became extinct in which era? (a) Mesozoic (b) Triassic (c) Cenozoic (d) Cretaceous
68. Lactose is derived from which products? (a) sugarcane (b) milk (c) grape fruit (d) ripe fruit
69. Insulin secretions is controlled by ___ and ___ (a) Alpha and beta cells (b) Pancreatic and intestinal juice (c) Glucose and gastric juice (d) Hyperglycemia and parasympathetic nervous system
70. ___ is a precursor of vitamin D found in plants and animals (a) ketone (b) phospholipids (c) rickets (d) sterols

1. C
2. B
3. A
4. A

5. D

7. B

8. A

10. A

11. A

12. A

13. C

14. B

15. A

16. B

17. A

18. B

Prokaryotic cells lack an organized and membrane-bound organelles. Their DNA is found in a central part of the cell called the nucleus. Structural proteins form the structural framework of various organs and connective tissues like hairs, nails, bones, cartilage, and skin. Examples of structural protein include; keratins collagens and elastins.

6. A

The nutrition of monera which can also be called Protista can either be any of the following, Autotrophic, Heterotrophic, Parasitic, Symbiotic, Saprophytic, etc.

9. B

Carrier proteins can transport ions (charged particles) and molecules either passively through facilitated diffusion or via secondary active transport while channel proteins only transport substances through membrane passively. Symporter is moving glucose up its concentration gradient (offer uphill movement).

It involves the oxidation-reduction action that transforms 6 molecules of water (H_2O) and 6 molecules of carbon dioxide (CO_2) into one glucose molecule and 6 molecules of O_2 .
 $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Darwinian natural selection was combined with Mendelian inheritance to form the modern evolutionary synthesis, in which the connection between the units of evolution (genes) and the mechanism of evolution (natural selection) was made.

If breastfeeding continues for longer than 6 months without an infant receiving solid food, the risk of marasmus can also increase, especially if the mother is malnourished herself.

DNA is made of chemical building blocks called nucleotides and this building blocks are made of 3 parts; a phosphate group, a sugar group and one of four types of nitrogen bases found in nucleotides which are: Adenine (A), Thymine (T), guanine (G) and Cytosine (C).

Neo Darwinism is a theory of evolution that is a synthesis of Darwin's theory in terms of natural selection and modern population genetics.

19. B
22. A

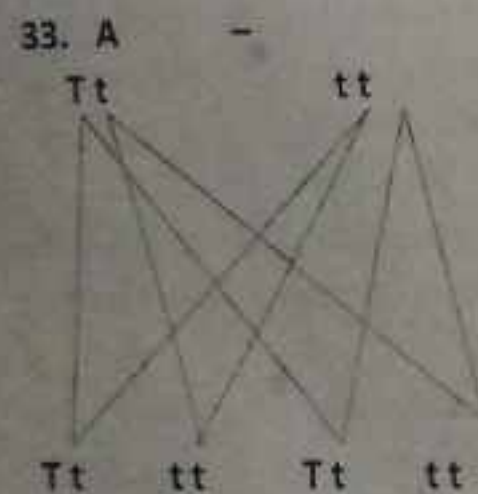
20. C

21. C

The centriole is a pair of cellular organelle that is adjacent to the nucleus and functions in the formation of spindle apparatus (consists of cylinder wit 9 microtubules) during cell division.
Amino acids are generally soluble in water and insoluble in non-polar organic solvents such as hydro-carbons.
Autotrophic nutrition is the type of nutrition whereby green plants manufacture their own food through the process of photosynthesis.

Viral cell can only exist as a living thing when it is found inside a living vessel.
Archeans and Eubacteria; they lack true nucleus or membrane bound nucleus.
3 Groups; monosaccharides, oligosaccharides and polysaccharides

Oxidation metabolism; it takes place in the mitochondrial matrix at the internal folded mitochondrial membranes (cristae).
Leptotene; This is the first stage of the prophase of meiosis during which each chromosome becomes visible as two fine threads (chromatids).
The nuclear membrane appears is the odd one out. In Diakinesis which is the final step of prophase 1 and it is the termination of condensing of the chromosomes. Nucleus disappears and the nuclear envelope disintegrates and the centriole move to the equator.
2 Tall: 2 Short



34. B

35. C

36. A

37. A

38. D

Africa: Homo erectus were the first of the Hominis to emigrate from 1.8 to 1.3 million years ago and Homo sapiens (modern man) first lived in Africa.
Carboxyl group: Amino acid contains alpha-amino carboxylic acid. Each molecule contains a central carbon (C) atom called X-carbon, to which both an amino acid and a carboxyl group are attached.

39. A

40. D

41. D

42. C

43. C

44. A

45. B

48. C

51. D

52. A

53. A

54. B

55. B

56. B

57. C

58. C

Essential amino acids: only 9 of the 20 amino acids are classified as essential because they can't be made by your body and must be obtained through your diet.
70S Units: The Eukaryotic ribosomes are also known as 80s ribosomes, referring to their sedimentation coefficient in Svedberg units, because they sediment faster than the prokaryotic (70s) ribosomes.

Replication: through the generation of abundant copies of its genome and packaging these copies, the virus continues infecting new hosts.

Half the number of chromosomes in the somatic cells.

Nucleus, mitochondria, chloroplast.
Double membrane bound organelles include: 1.) Mitochondria 2) Chloroplasts 3) Endoplasmic reticulum 4) Golgi body

46. D

47. B

49. C

50. B

Glycogen is a peptide hormone, produced by alpha cells of the pancreas, it works to raise the concentration of glucose and fatty acids in the bloodstream by breaking down glycogen into glucose in the liver.

Species

Energy can be transformed from one form to another.

Entropy in the universe is increasing: The second law of thermodynamics states that the total entropy of an isolated system can never decrease overtime or it can be increasing if it is not in an isolated system.

Metabolism which includes both catabolism and anabolism

From $\Delta G = \Delta H - T\Delta S$
Where ΔG = change in free energy; ΔH = change in enthalpy; T = absolute temperature; ΔS = change in entropy
 $\Rightarrow \Delta H = \Delta G + T\Delta S$
If ΔG is negative, ΔH will also be negative.
Therefore, the reaction is exothermic.

ΔH is negative

Since glucose leads to two triose sugars in preparatory phase, each reaction in the pay-off phase occurs twice per glucose molecule. This

yields 2 NaOH molecules and 4 ATP molecules, leading to a net gain of 2 ATP molecules from glycolytic pathway per glucose.

59. C

H⁺ combines with O₂ inside the mitochondria creating water (H₂O). In most Eukaryotic cells, 36 ATP are produced from 1 glucose molecule: 2 from Glycolysis and 2 from the citric acid cycle by substrate level phosphorylation and 32 from Electron transport by oxidative phosphorylation.

60. B

The mitotic spindle appears during the early prophase and it is made up of microtubules, strong fibres that are part of cells skeleton.

61. A

62. B

Equational division and it is analogous to mitosis, in which the sister chromatids are segregated, creating 4 haploid daughter cells. The meiosis I is referred to as reductional division because the diploid to haploid change.

63. C

Zygote is the sub-stage where synapsis between homologous chromosomes begins. It is also known as zygonema.

64. B

65. A

66. A

67. D

Dinosaurs became extinct about 65 million years ago at the end of cretaceous period.

68. B

69. A

70. D

FIRST SEMESTER EXAMINATION 2018/2019 SESSION
FSB 101.1 (GENERAL BIOLOGY 1)

DURATION: 1HR

INSTRUCTION: ANSWER ALL QUESTIONS

- About _____ million species have been studied and classified (a) 2 (b) 100 (c) 5 (d) 1
- In the hierarchy of biological classification directly BELOW the family is the _____ (a) Species (b) Phylum (c) Order (d) Genus
- Macroevolution is _____ of micro evolution (a) Accumulation (b) Subtraction (c) Configuration (d) Equal to
- Evidence suggests that life on earth has existed for about _____ billion years (a) 4.6 (b) 7 (c) 3.7 (d) 5
- One of the fathers of evolution is (a) Linnaeus (b) Malthus (c) Oparin (d) Curry
- Who supported the idea of inheritance of acquired characteristics (a) Count Buffon (b) Baptiste Lamarck (c) Charles Darwin (d) Stanley Millers
- One of the importance of evolution is (a) It shows feeding habits (b) Shows mode of locomotion (c) Shows flexibility (d) Shows shared ancestry
- Current ideas on evolution are usually referred to as (a) Survival of the fittest (b) Natural selection (c) Selection by descent (d) Modern synthesis
- The scientist that came up with modern synthesis is _____ (a) Futuyama (b) Alfred Russell (c) Charles Darwin (d) Jean Baptiste
- Importance of classification include the following except _____ (a) Placing organisms in groups (b) Makes organism study easier (c) Show relationship of different organisms (d) Show how to preserve organisms
- A post anal tail found vertebrates called _____ (a) Coccyx (b) Notochord (c) Mantle (d) Endostyle
- The locomotory structure of _____ is by cytoplasm (a) Sporozoan (b) Ciliophora (c) Sarcodinian (d) Pseudopodia
- The fathers of natural selection are _____ and _____ (a) Carolos Linnaeus and Charles Darwin (b) Georges Cuvier and Lamarck (c) Alfred Russell and Charles Darwin (d) Alfred Russell and Count Buffon
- The proliferation of the jewels fishes was during _____ period (a) Silurian (b) Devonian (c) Carboniferous (d) Ordovician
- Monotremes appeared in which period _____ (a) Cenozoic (b) Jurassic (c) Cretaceous (d) Mesozoic
- Stomatolits were discovered at _____ (a) Lake Tanganyika (b) Olduvai Gorge (c) Lake Superior (d) Lake Australia
- Nuclear envelope is absent in _____ (a) Protozoa (b) Bacteria (c) Onion (d) Amoeba
- An antihemorrhagic vitamin is a found in the intestine of a _____ person (a) Sick (b) Healthy (c) Malnourished (d) Bleeding
- Imbalance in dietary intake is called (a) Deficiency (b) Disease (c) Marasmus (d) Malnutrition

20. Another name of cholecalciferol is ☒ (a) Vitamin D (b) Vitamin E (c) Vitamin K (d) Vitamin B
21. A combination of glucose and fructose gives (a) Milk sugar ☒ (b) Malt sugar (c) Ovaltine sugar (d) Table sugar
22. _____ is a German scientist that coined the term ecology (a) Lamarck in 1866 (b) Haeckel in 1869 (c) Lamarck in 1869 ☒ (d) Haeckel 1866
23. The ability to interbreed and produce offspring is called _____ (a) Biotic resistance (b) Environmental resistance ☒ (c) Biotic potential (d) Species
24. The functional role of an organism is regarded as (a) Community (b) Ecosystem ☒ (c) Niche (d) Tropic
25. An ecosystem that covers a wide geographical area is called (a) Community ☒ (b) Biome (c) Ecotone (d) Biosphere
26. A relationship that negative-negative (-/-) is called (a) Competition (b) Neutralism (c) Parasitism ☒ (d) Saprophytism
27. An example of cohesion phenomenon is _____ (a) Metabolism (b) Anabolism (c) Vaporization ☒ (d) Surface tension
28. _____ exhibit both adhesion and cohesion behaviour in a plant life (a) Glass (b) Metal ☒ (c) Water (d) Air
29. Carbohydrates comprise only of ☒ (a) O, C, H (b) Na, C, H (c) O, C, Zn (d) O, H, Fe
30. An example of acidic amino acid is _____ (a) Alanine (b) Tyrosine (c) Glutamic (d) Arginine
31. An example of simple triglycerides is ☒ (a) Tristearin (b) Beeswax (c) Dioleopalmitin (d) Carnauba
32. _____ discovered purine and pyrimidine basis of nucleic acids (a) Friedrich Miescher (1869) (b) Fischer (1980) ☒ (c) Zacharis (1881) (d) Hertwing (1884)
33. One of the major characteristics of RNA is to _____ (a) Carry genetic information called genes (b) Made of two chains made of polymer unit nucleotides ☒ (c) Convert genetic information from genes into amino acid sequences of proteins (d) The prokaryotes store the DNA in the cytoplasm
34. In bacterial cell, flagella covering the entire surface of the cell is _____ (a) Monotrichous (b) lophotrichous ☒ (c) Petritrichous (d) Amphitrichous
35. The following factors influence photosynthesis except _____ (a) Light (b) Temperature (c) CO₂ ☒ (d) Oil
36. Light reaction takes place in the ☒ (a) Stroma (b) cell (c) thylakoid (d) mitochondria
37. Malaria is caused by _____ (a) trypanosoma ☒ (b) mosquito (c) plasmodium (d) Rhizobium
38. Which of these options is correct order. (a) Molecule > element > organ > organism (b) Atom > element > molecule > compound ☒ (c) Atom > element > compound > molecule (d) Atom > element > molecule > compound.
39. Formation of molecules is either by sharing or transfer of electrons between atoms. Among these options indicate the molecules with ionic bond (a) Sodium and oxygen ☒ (b) Nitrogen and oxygen (c) Phosphorus and chlorine (d) Calcium and chalk

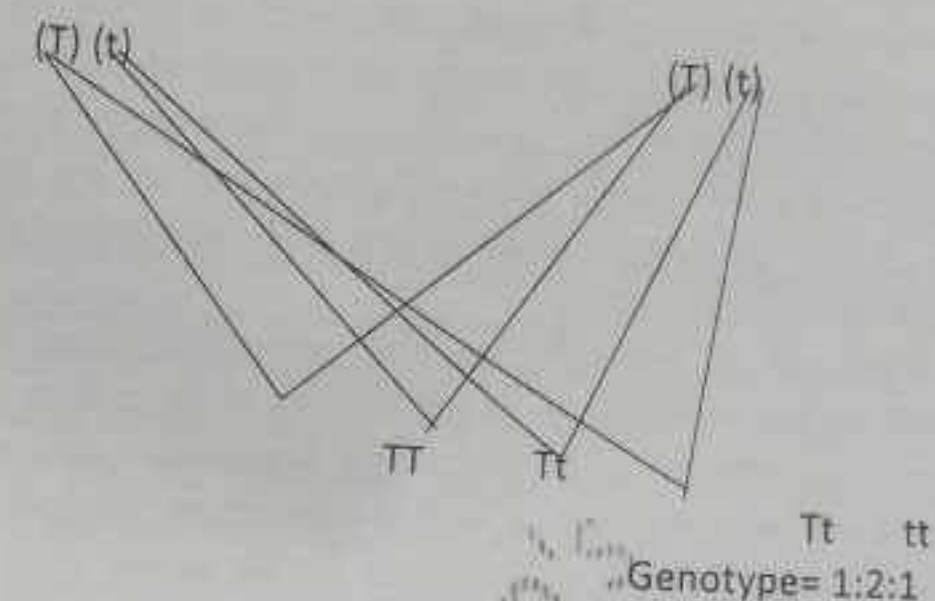
40. The following are properties of water except ☒ (a) Miscibility of oil (b) Polar molecule (c) Has high specific heat capacity (d) High latent heat of vaporization
41. What role do buffer solution play in biological systems? ☒ (a) None (b) Role yet to be defined (c) Control the pH of biological systems (d) Control the temperature of blood
42. Cellular energy comes mainly from (a) Inorganic compounds (b) Lean meat (c) Deoxyribonucleic acid ☒ (d) Carbohydrates
43. Homopolysaccharides are complex carbohydrates that yield the _____ types of monomers on breakdown (a) Different (b) Combination (c) No know ☒ (d) Same
44. Glycogen and starch are two ways organisms store excess carbohydrates in their bodies. Indicate the correct option (a) Glycogen for plants (b) Starch for humans ☒ (c) Glycogen for animal (d) Starch for animals (e) Both glycogen and starch for higher plants.
45. Why do lipids store more energy than carbohydrates? (a) Are close to water (b) Have short carbon chains ☒ (c) Longer carbon chains than carbohydrates and proteins (d) Are greasy
46. The central dogma of cell biology states that information flows from _____ (a) Water to molecules then to atoms (b) From carbohydrates and proteins (c) Proteins to lipids then to carbohydrates ☒ (d) Deoxyribonucleic acid to messenger ribonucleic acid then to protein
47. Osmosis is defined as _____ (a) Movement of materials from a region of high to a region of low concentration (b) Movement of materials downhill ☒ (c) Movement of water from a region of high to that of low concentration through a partially permeable membrane (d) Any movement of materials involving expenditure of energy
48. Passive transport into cells require all these except _____ ☒ (a) Energy from cells (b) involves water and small materials (c) Partially permeable membrane (d) Very large molecules
49. Homopolysaccharides are complex carbohydrates that yield the _____ types of monomers on breakdown. (a) Different (b) Combination (c) No known ☒ (d) Same
50. Exosmosis in extreme cases results in _____ (a) Endosmosis (b) Osmosis ☒ (c) Endocytosis (d) Plasmolysis
51. The difference between osmosis and dialysis is _____ (a) Both involve diffusion (b) Materials move from region of high concentration to that of low concentration (c) No movement of materials ☒ (d) Water is move in osmosis but solutes are moved in dialysis
52. Large materials are actively transported across partially permeable membranes through carriers provide the option with the right types of carriers in active transport of materials (a) Membraneport, activeport and inactiveport ☒ (b) Symport, Uniport and Antiport (c) Sympart, Unipart, Antipart (d) Membraneport, Uniport and Activeport
53. Phagocytosis and Pinocytosis can be defined as _____ and _____ (a) Movement of materials into and out of from cells (b) Movement of materials in and out from living organisms (c) Cell drinking and cell eating (d) Cell eating and drinking
54. Bioenergetics is the study of (a) Energy transformations in a physical system (b) Energy transformations in a physicochemical system (c) Energy transformation in a biological system ☒ (d) Energy transformation in an adiabatic
55. Which of the following statements is NOT correct (a) Thermodynamics tells us whether a reaction will occur spontaneously (b) Energy is used to do work (c) Radiant energy can be converted to thermal energy ☒ (d) Energy is not used for reproduction

56. Which of the following statement is NOT true for the second law of thermodynamics? (a) Energy is lost as disorder increases (b) Entropy involves the transformation of potential energy into heat energy (c) Entropy in the universe is continuously increasing (d) Energy is gained as entropy increases
57. When the net change in entropy of reaction is positive, it means that (a) The heat energy is increased (b) The entropy is increased (c) The heat energy is decreased (d) The entropy is decreased
58. Catabolic reactions are examples of (a) Exothermic reactions (b) Endothermic reactions (c) Exergonic reactions (d) Endogenic reactions
59. Which of these statements is correct? (a) Enzymes reduce the activation energy required to start a biochemical reaction (b) Activation energy increases the energy required to start a biochemical reaction (c) The activation energy in biochemical reactions are low in the absence of enzymes (d) Enzymes are not organic catalyst
60. Which of the following features is absent in metaphase in mitosis (a) Appearance of chromatids (b) Appearance of spindle (c) Appearance of the centromere (d) Appearance of nucleus
61. Which of the following stages in meiosis I does the nuclear membrane begin to form around the chromosomes at the two poles (a) Anaphase I (b) Metaphase I (c) Telophase I (d) Prophase I
62. The gametes from the cross between two individuals with the genetic constitution SS and ss where S is the allele for smooth skin and s is the allele for wrinkled skin are (a) S and s (b) SS and ss (c) Ss and Ss (d) s and s
63. The phenotype of an individual genotype Tt where T is the allele for tall and is dominant over recessive allele t for short stem (a) Short stem (b) Tall stem (c) Medium height for stem (d) All of the above
64. What are the possible genotypes of the cross between two heterozygous individuals (Tt and Tt) where T is the allele for tall stem and it is dominant over the allele t for short stem (a) TT, Tt and tt (b) TT and tt (c) Tt and tt (d) TT and Tt
65. Which of the following expression is correct? Gibbs free energy = ΔG ; Enthalpy = ΔH ; Entropy = ΔS (a) $\Delta G = -\Delta H + T\Delta S$ (b) $\Delta G = \Delta H + T\Delta S$ (c) $\Delta G = \Delta H - T\Delta S$ (d) $\Delta G = \Delta H - T\Delta S$
66. Two important organelles involved in energy transformation are (a) Mitochondria and ribosomes (b) Mitochondria and chloroplast (c) Chloroplast and ribosome (d) None of the above
67. In a monohybrid cross between pure breed black goat and breed white (recessive) goat, what percentage of the offspring are likely to be black? (a) 0 (b) 20 (c) 50 (d) 100
68. Amitosis is the cell division without nuclear envelope breakdown and formation of spindle fibre (a) True (b) False (c) All of the above (d) None of the above
69. Synapsis takes place during (a) Prophase (b) Prophase I (c) Metaphase (d) None of the above
70. Prophase I in meiosis consists of the following except (a) Leptotene (b) Zygotene (c) pachytene (d) Prometaphase

1. B There about 1.5 million species that has be successfully classified.
2. D Kingdom → Phylum → Order → Family → genus → Species
3. A
4. A
5. A
6. C Jean-Baptiste Lamarck corned the theory that acquired characteristics can be inherited and Charles Darwin supported the theory.
7. D This shows shared ancestry
8. D Modern synthesis: This theory seems to reconcile Darwin's theory of evolution and Mendel Gregor idea of hereditary.
9. D Modern synthesis is described by Futuyama in 1986
10. D Shows how to preserve organism.
11. A Coccyx is the tail bone.
12. D Shows how to preserve organism.
13. C
14. D
15. B
16. D Lake Africa. Stomatolites were the first and oldest fossil that shows the evidence of life on earth.
17. B
18. D Bleeding
This is an agent that promotes hemostasis (stop bleeding)
19. B
20. A Vitamin D. Cholecalciferols is a type of Vitamin D which is made by the sun when exposed to sunlight.
21. B Malt Sugar. Malt sugar is the combination of Glucose and Fructose.
22. D Haeckel. Ernst Haeckel corned the world ecology in the year 1866 from the Greece word "Oikrs"-house and "logos" meaning study.
23. C Biotic Potential. This is the ability to interbreed and produce offspring (species)
24. C Niche: is defined as a fundamental role of an organism

25. B Biome: is an ecosystem that covers a wide geographical area
26. D Saprophytism: The ability of certain plant to live in dead or decaying matter
27. D 28. C
29. A O, C, H. Carbohydrates comprises of Oxygen, Carbon and hydrogen.
30. C Glutamine. These have active sili chain at neutral P
31. A 32. C
33. C It converts genetic information from gene to amino acid sequence of proteins.
34. C Peritrichous. Numerous flagella all over the bacterial body.
35. D Oil. Oil does not influence photosynthetic process.
36. C Stroma. Light dependent reactions take place on the stroma outside of the thylakoid membrane.
37. B Mosquito. Mosquito is the vector that carry the disease malaria.
38. C 39. C
40. C Miscibility with oil. Water is not miscible with oil rather it is immiscible.
41. C
42. D Carbohydrate. Carbohydrate is an energy giving food. It provides energy for cellular activities.
43. D 44. C 45. C 46. D
47. C Osmosis. This is the movement of water molecules from a region of high concentration to a region of low concentration through a semi-permeable membrane.
48. A 49. D
50. D Plasmolysis. Plasmolysis occurs in extreme water loss.
51. D 52. B
53. D Cell eating and drinking. Phagocytosis is referred to as cell-eating while Pinocytosis refers to cell drinking.
54. C 55. D

56. A Energy is lost as disorder increase. Energy is the ability to do work while entropy is the rate of disorderliness in a system.
57. D 58. B
59. A Enzymes Reduces the activation. Energy required to start a biochemical reaction.
60. A 61. B
62. A



Meaning - 1 tall stem, 2 heterogeneous tall stem (medium size) 1 short stem.
But the phenotype = 3:1
Meaning - 3 tall stem and 1 short stem

64. A 65. C 66. B 67. C
68. A True. A mitosis is a cell division by simple cleavage of the nucleus and division of the cytoplasm with spindle formation or appearance of chromosome.
68. D None of the above. Synapsis takes place during meiosis which is referred to as the pairing of two chromosomes that occurs during meiosis.
70. D

FIRST SEMESTER EXAMINATION 2019/2020 SESSION

FSB 101.1: General Biology 1 Time: 1:30mins.

Instruction: Answer All Questions

- Cellular energy comes mainly from (a) Inorganic compounds (b) Lean meat (c) Deoxyribonucleic Acid (d) Phosphoproteins (e) Carbohydrates
- Heteropolysaccharides are complex carbohydrates that yield the ___ types of monomers on Breakdown (a) Different (b) Similar (c) No known (d) Five (e) All the above
- Glycogen is the form of stored carbohydrates in which organisms? (a) Plants (b) Starch for humans (c) Animals and plants (d) Animals (e) None higher plants
- Why do lipids store more energy than carbohydrates? (a) Are close to water (b) Have short carbon chains (c) Longer carbon chains than carbohydrates and proteins (d) Are greasy (e) Humans can drink lipids
- The central dogma of cell biology states that information flows from ___ (a) Water to molecules then to atoms (b) From carbohydrates to lipids then to proteins (c) Proteins to lipids then to carbohydrates (d) Protein to deoxyribonucleic acid then to messenger ribonucleic acid (e) Deoxyribonucleic acid to messenger ribonucleic acid then to protein
- A good quality dietary protein should have (a) No amino acids (b) Limited amount of amino acids (c) All the essential amino acids in sufficient amounts (d) Combination of oil and milk (e) Bones
- Osmosis is defined as ___ (a) Movement of materials from a region of high to a region of low concentration (b) Movement of materials downhill (c) Movement of water from a region of high to that of low concentration through a partially permeable membrane (d) Any movement of materials involving expenditure of energy (e) Movement against an energy gradient
- Passive transport into cells will require all these except ___ (a) Energy from cells (b) Involves water and small materials (c) Partially permeable membrane (d) Very large molecules (e) Include osmosis, diffusion and facilitated osmosis
- Lipids have hydrophilic tails and hydrophobic heads (a) True (b) False (c) Not always (d) Statement is true for proteins (e) None of the above
- Exosmosis in extreme cases result in ___ (a) Endosmosis (b) Osmosis (c) Endocytosis (d) Plasmolysis (e) Plantasmolysis
- If purple flower color is dominant to white color and these are crossed, what proportion of the offspring will be purple in the F1 generation? (a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) None (d) 20 (e) All
- Which of these is not an algal group? (a) Mycota (b) Platyhelminthes (c) Escherichia (d) Mangifera indica (e) All of the above
- Chromosomes are made up of genes which are in turn made up of ___ strands (a) RNA (b) Nucleus (c) Cytoplasm (d) DNA (e) Cartilage
- The vegetative bodies of fungi comprise of strands called hyphae which associate to ___ (a) Filaments (b) Whatever (c) Mycellia (d) Spores (e) Sporangia
- Phagocytosis can be defined as (a) Movement of materials into cells (b) Movement of materials into living organisms (c) Cell drinking (d) Cell eating (e) All the above
- The following are characteristics of prokaryotic cells except the (a) Presence of membrane bounded organelles (b) Possession of ribosomes (c) Absence of membrane bounded nucleus (d) Membrane bounded organelles (e) Absence of microtubules and microfilaments

- Which of the following statements is not true (a) Archaea are found in extreme environments (b) The cell walls of eubacteria contain peptidoglycan (c) The concentration of peptidoglycan in Gram negative bacteria is much lower than in Gram positive bacteria (d) Gram positive bacteria stain red or pink in the Gram stain procedure (e) Archaea contain methanogens
- Which of the following statements is not true? (a) The cell is the fundamental unit of structure and function in a living organism (b) Viruses are unicellular organisms (c) Differentiation and specialization of cells has led to the diversity and complexity of different types of organisms (d) Plants and animals are multicellular organisms (e) Bacteria are unicellular organisms
- The following structures are present in eukaryotic cells except? (a) Nuclear membrane (b) Mitochondria (c) Mesosomes (d) Chloroplast (e) Chromosomes
- The following statements are correct for the cytoplasmic membrane of prokaryotic cells except (a) It consist of a bilayer of phospholipid sandwiching a protein layer (b) The enzymes responsible for glycolysis are situated in the cell membrane (c) It is thrown into folds called mesosomes (d) The enzymes responsible for the Krebs Cycle and the electron transport chain are located in the cell membrane (e) Enzymes responsible for the Light reaction of photosynthesis are located in the cell membrane
- The functions of the pili (fimbriae) include all of the following but (a) used for attachment by eukaryotic cells (b) used for attachment by prokaryotic cells (c) can facilitate pathogenicity in bacteria (d) enables the formation of conjugation tubes in bacteria (e) facilitates exchange of plasmids between bacteria cells
- Which statement is not true for the cell wall of eubacteria (a) gives the cell shape (b) provides a rigid structural support for the cells (c) Protects the cells from harsh environments (d) site for Gram reaction (e) Mycoplasma are eubacteria that do not possess cell walls
- Eukaryotic cells possess the following organelles except? (a) Mitochondria (b) Plastids (c) Lysosomes (d) Nuclear body (e) Golgi body
- The following statements are true for the endoplasmic reticulum (ER) except? (a) a site for detoxification of toxins (b) a site for export of manufactured carbohydrates (c) site for the translation stage of protein synthesis (d) Rough and Smooth ER exist (e) site for export of proteins
- The following statements are true for mitochondria except? (a) The Krebs cycle enzymes are located in the mitochondria (b) The electron transport chain enzymes are located in the mitochondria (c) The enzymes responsible for glycolysis are located in the mitochondria (d) Bacteria do not possess mitochondria (e) The mitochondria possess inner and outer membranes
- The following statements are true for ribosomes except (a) Structurally they are composed of two fragments (b) They are the sites for translation stage of protein synthesis (c) They are the stage for the transcription stage of protein synthesis (d) some ribosomes are attached to the endoplasmic reticulum (e) mitochondria and chloroplast possess ribosomes
- Enzymes responsible for hydrolysis in the cells are contained in (a) Peroxisomes (b) Lysosomes (c) Autosomes (d) Ribosomes (e) Chromosomes

28. The following statements are true except (a) Centrioles (b) The spindle consists of an array of microtubules (c) The flagella and cilia of eukaryotes are composed of microtubules (d) The flagella of eukaryotic cells are composed of microtubules (e) Centrioles do not play roles in meiosis
29. The following statements are true of the nucleus of eukaryotic cells except (a) are surrounded by nuclear membrane (b) Contain chromosomes (c) contain a nucleolus consisting of RNA (d) contains DNA (e) are not surrounded by a nuclear membrane
30. Kinetic energy is the (a) ability to do work (b) is the energy of motion (c) is stored up energy (d) is the energy in a glucose molecule (e) energy in a concentration gradient
31. Which of the following is not correct (a) An atom or molecule that loses an electron is said to be oxidized (b) An atom or molecule that gains an electron is said to be reduced (c) The reduced form of a molecule has a higher level of energy than the oxidized form (d) The reduced form of a molecule has a lower level of energy than the oxidized form (e) Oxidation and reduction always takes place together
32. Energy flows through the ecosystem in one direction from the sun through (a) Photosynthetic organism to herbivores, carnivores and decomposers (b) algae, herbivores, carnivores and decomposers (c) green plants, herbivores, carnivores and heterotrophic bacteria (d) photosynthetic bacteria, herbivores and carnivores (e) photosynthetic plant, herbivore and carnivores
33. All consumers are examples of (a) autotrophic organisms (b) holozoic organisms (c) heterotrophic organisms (d) chemoautotrophic organisms (e) photoautotrophic organisms
34. Bioenergetics is the study of (a) energy transformation in a closed system (b) energy conversion in a biochemical system (c) energy conversion in a biological system (d) energy transformation in an open system (e) energy transformation in an isolated system
35. Which of the following statements is not correct for the first law of thermodynamics (a) Energy cannot be created or destroyed it can only change from one form to another (b) Energy can change from potential energy to chemical energy (c) The total amount of energy in an open system is constant (d) The total amount of energy in the universe is a constant (e) There is exchange of energy and matter in a closed system
36. The following statements are correct for the second law of thermodynamics (a) The disorder in the universe is increasing (b) The entropy in the universe is increasing (c) Disorder is more likely to occur than order (d) Energy transformation proceed spontaneously to convert matter from a more ordered, less stable form to a less ordered but more stable form (e) Disorder is less likely to occur than order
37. The net effect, amount of energy actually available to break and subsequently form other Chemical bonds is called (a) Enthalpy (b) Entropy (c) Gibbs free energy (d) Exergonic Energy (e) Endergonic Energy
38. The enthalpy when the chemical reaction release heat is (a) ΔH (b) $-\Delta H$ (c) $-\Delta S$ (d) ΔS (e) ΔG
39. When the products of a reaction are less complex and more disordered than the reactants, the reaction proceeds with a (a) $-\Delta S$ (b) ΔH (c) ΔG (d) ΔS (e) $-H$
40. When the products of a reaction are more complex and less disordered than the reactants, the reaction proceeds with a (a) $-\Delta S$ (b) ΔH (c) ΔG (d) ΔS (e) $-\Delta H$
41. In a spontaneous reaction (a) ΔG (b) ΔS (c) $-\Delta G$ (d) ΔH (e) endergonic reaction
42. The breakdown of glucose into pyruvic acid is an example of a (a) anabolic reaction (b) catabolic reaction (c) endergonic reaction (d) heptagonic reaction (e) oragonic reaction
43. The following statements are correct for enzymes except (a) Enzymes are specific for the substrates they catalyze (b) The active site is located on the enzyme (c) The active site is located on the substrate (d) The enzyme can be reused (e) The enzyme lower the activation energy for the reaction to start
44. When the change in free energy is positive the reaction is (a) exergonic (b) spontaneous (c) endergonic (d) amphibolic (e) anaphlerotic
45. The following statements is correct for glycolysis except (a) 2ATP molecules are used up (b) 2NADH molecules are produced (c) 4ATP molecules are produced (d) There is a net production of 2ATP molecules (e) Two ATP molecules are produced
46. Conjugated proteins are also known as (a) complex proteins (b) Holoproteins (c) Protamines (d) Histones (e) Quaternary proteins
47. Carbohydrates are saccharides and are grouped into (a) 3 groups (b) 2 groups (c) 5 groups (d) 6 groups (e) 7 groups
48. Amino acids are _ _ _ (a) insoluble in water (b) only carboxyl compounds (c) Building blocks of proteins (d) None of the above (e) only amino groups
49. Based on shape, proteins are classified into _ _ _ and Fibrous (a) Globular (b) Triangular (c) Hexagon (d) Enzymes (e) Fibrilla
50. Simple proteins are also known as (a) Kinetprotein (b) Ctoproteins (c) Mega proteins (d) Haloproteins (e) Heteroproteins
51. Nucleic acids are of two types (a) DNA and BNA (b) BNA and RNA (c) RNA only (d) DNA and RNA (e) DNA and RDA
52. DNA is also known as (a) Dexyribonucleic acid (b) Oxyribonucleic acid (c) Deoxyribonucleic acid (d) Oxy nucleic acid (e) Denitronucleic acid
53. Attraction between molecules of different substances as seen of water molecules is known as (a) Cohesion (b) Adhesion (c) Gumming (d) All of the above (e) None of the above
54. Communication between cells and in cells occur in how many ways? (a) 1 (b) 2 (c) 3 (d) 4 (e) 5
55. Osmosis is _ _ _ (a) Active transport (b) Passive transport (c) Exocytosis (d) Phagocytosis (e) Picocytosis
56. The transportation of materials that involves ions and charged particles in the cell is called (a) endocytosis and Exocytosis (b) Active Transport (c) Passive Transport (d) None of the above (e) Phagocytosis
57. Bulk materials can be transported in the cell through (a) Osmosis (b) Endocytosis and Endocytosis (c) Passive Transport (d) None of the above (e) Active Transport
58. One of these is not a type of evolution (a) Polymerization (b) Natural Selection (c) Macro evolution (d) Micro evolution (e) Large scale evolution
59. The Neo Darwin theory of evolution (a) Recognizes symbiosis (b) recognizes natural selection (c) Fixed and provided evidence of decent (d) recognizes the importance of mutation and variation (e) All of the above
60. The modern man was first discovered in (a) Asia (b) Australia (c) Africa (d) Europe (e) America

61. One of these is not an evidence of evolution (a) Gene variation (b) Fossil records (c) Embryological records (d) Cell division (e) Ribosomal RNA sequencing
62. Prokaryotic cell can only divide through one of these (a) Binary fission (b) Mitosis (c) Cytoplasmic streaming (d) None of the above (e) Meiosis
63. The cell division that occurs in somatic cells (a) Binary fission (b) Interphase (c) Meiosis (d) Mitosis (e) Sporulation
64. Meiosis is only seen (a) Anaphase (b) Ureter (c) in the kidney (d) in the gonads (e) somatic cells
65. The first phase of cell division is called? (a) cell cycle (b) metaphase (c) telophase (d) prophase (e) anaphase
66. The homologous pair of chromosomes are joined at (a) centromere (b) chiasmata (c) nucleus (d) Chromatids (e) Chromatin
67. One of the following is a way of sexual reproduction in protozoa (a) Transformation (b) Consummation (c) Transduction (d) Conjugation (e) Transfection
68. Amitosis could be called any of the following except? (a) Simple cell division (b) Direct cell division (c) Closed cell division (d) Mitosis (e) None of the above
69. In mitosis chromosomes replicate into the diploid state (a) True (b) False (c) True when daughter cells are produced (d) This only happens in metaphase (e) None of the above
70. Karyokinesis is subdivided into ___ phases (a) 1 (b) 2 (c) 3 (d) 4 (e) 5

Johnson's

12 15

11 : 37

2 : 63

14-00

1. E
2. E
3. D
4. C
5. E
6. C
7. C
8. D
9. A
10. D
11. E
12. E
13. D
14. C
15. E
16. C
- 17.
18. B
19. C
- 20.
21. E
22. C
23. All of the above
24. C
25. C
26. C
27. B
28. E
29. E
30. B
31. D
32. A
33. C
34. C
- 35.
- 36.
37. A
38. A
39. D
40. C
41. A
42. B
43. C
44. A
45. D
46. C
47. C
48. C
49. A
50. D
51. D
52. C
53. B
54. D
55. B
56. B
57. E
58. A
59. E
60. C
61. E
62. A
63. D
64. D
65. D
66. A
67. D
68. B
69. A
70. D

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